

Variable-rate streaming with a spike-count readout

exp082 · 10 August 2026 · Draft

Abstract

This experiment tests whether PING networks trained across input rates can classify continuous MNIST streams with the same spiking readout used during training. Excitatory spikes drive ten output LIF neurons, and each class logit is the corresponding neuron’s total spike count within the current digit’s presentation window. Matched-condition and variable-condition streams show the online network response, while a 200 ms rate psychometric and a duration-by-rate map measure performance across the trained input distribution.

Methods

1. Training source

The experiment loads `ping_variable_rate_seed42`, `ping_variable_rate_seed43`, and `ping_variable_rate_seed44` from exp022. Each PING network is trained by sampling one maximum-pixel Poisson rate independently per image presentation from 0.5, 0.75, 1, 1.5, 2, 3, 5, 7.5, 10, 15, and 25 Hz. The recurrent weights and classifier are frozen throughout this experiment.

Note. The exp022 cells are registered and `tools/snn` implements the required output-LIF spike-count training, restore, and output-raster path. Presentation boundaries reset only the output LIF and its counter; the hidden PING state remains continuous. This entry still fails loudly if the variable-rate checkpoint bank is absent or uses another readout.

Streaming inference uses each seed’s checkpoint selected by validation accuracy, because this is a deployment evaluation of the selected classifier rather than an analysis of final-epoch drift.

2. Spike-count readout

At each timestep, the 1024-element excitatory spike vector $\mathbf{s}^{E(t)}$ drives a trained 1024-by-10 projection W_{out} into ten output LIF neurons. For a presentation beginning at timestep a and ending before timestep b , class evidence is

$$z_c = \sum_{t=a}^{b-1} s_c^{\text{out}(t)}, \quad c \in \{0, \dots, 9\}. \quad (1)$$

Here $s_c^{\text{out}(t)}$ is the spike emitted by output LIF neuron c at timestep t , and $\mathbf{z}$ contains the ten dimensionless output spike-count logits. The predicted digit is the index of the largest count. Dividing counts by the common presentation duration gives firing rates for reporting and leaves this argmax unchanged, but rates are not passed to cross-entropy as logits. The output

neurons therefore have membrane state, leak, threshold, spikes, and reset; unlike `mem-mean`, their membrane voltages are not the logits.

3. Matched inference

Every digit is read from only the spikes generated during its own presentation:

$$T_{\text{readout}} = T_{\text{presentation}}. \quad (2)$$

Here T_{readout} is the spike-summation window and $T_{\text{presentation}}$ is the duration of the corresponding digit. The first protocol holds both at 200 ms and presents a five-digit stream. This checks the trained duration and readout before either input rate or duration is varied.

4. Variable-rate and variable-duration inference

A second five-digit stream changes both presentation duration and maximum-pixel Poisson rate at digit boundaries. Each digit retains its own matched readout window. A factorial evaluation then crosses presentation durations 25, 50, 100, and 200 ms with the eleven training rates. Every cell is evaluated on 200 classified digits independently for seeds 42, 43, and 44. Independent five-digit streams are evaluated in simulator batches; each batch member has separate neuronal state, while hidden state remains continuous only between the five digits belonging to that stream. Statistical grid cells retain compact per-presentation spike counts and population totals. Full rasters are retained only for the two illustrative streams.

5. Psychometric protocol

The rate psychometric fixes presentation and readout duration at 200 ms, varies the maximum-pixel rate across the eleven training values, and reports held-out classification accuracy across the three trained seeds. This isolates rate robustness from shortened evidence windows.

Results

1. Matched-condition stream

The matched stream holds the maximum-pixel input rate at 5 Hz and gives every digit 200 ms of presentation and readout time. Vertical lines mark digit boundaries. The class traces are softmax transformations of cumulative output-LIF spike counts within the current digit; the counts reset at each boundary while the hidden PING state continues.

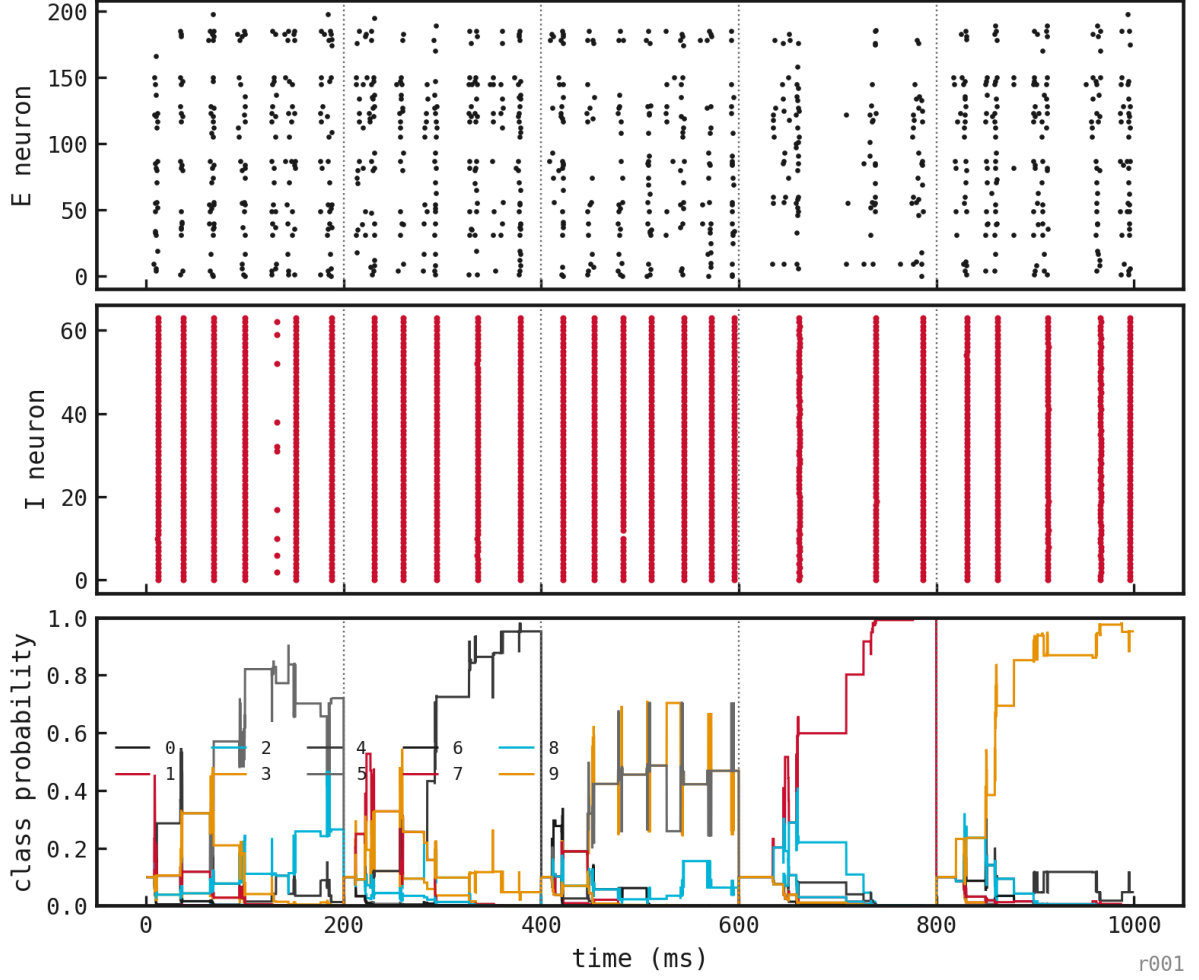


Figure 1: Matched-rate streaming inference with presentation duration equal to the spike-count readout window.

2. Variable conditions

The variable stream changes both input rate and presentation duration at digit boundaries. Each decision still uses exactly the spikes emitted during that digit, so evidence never leaks across labels.

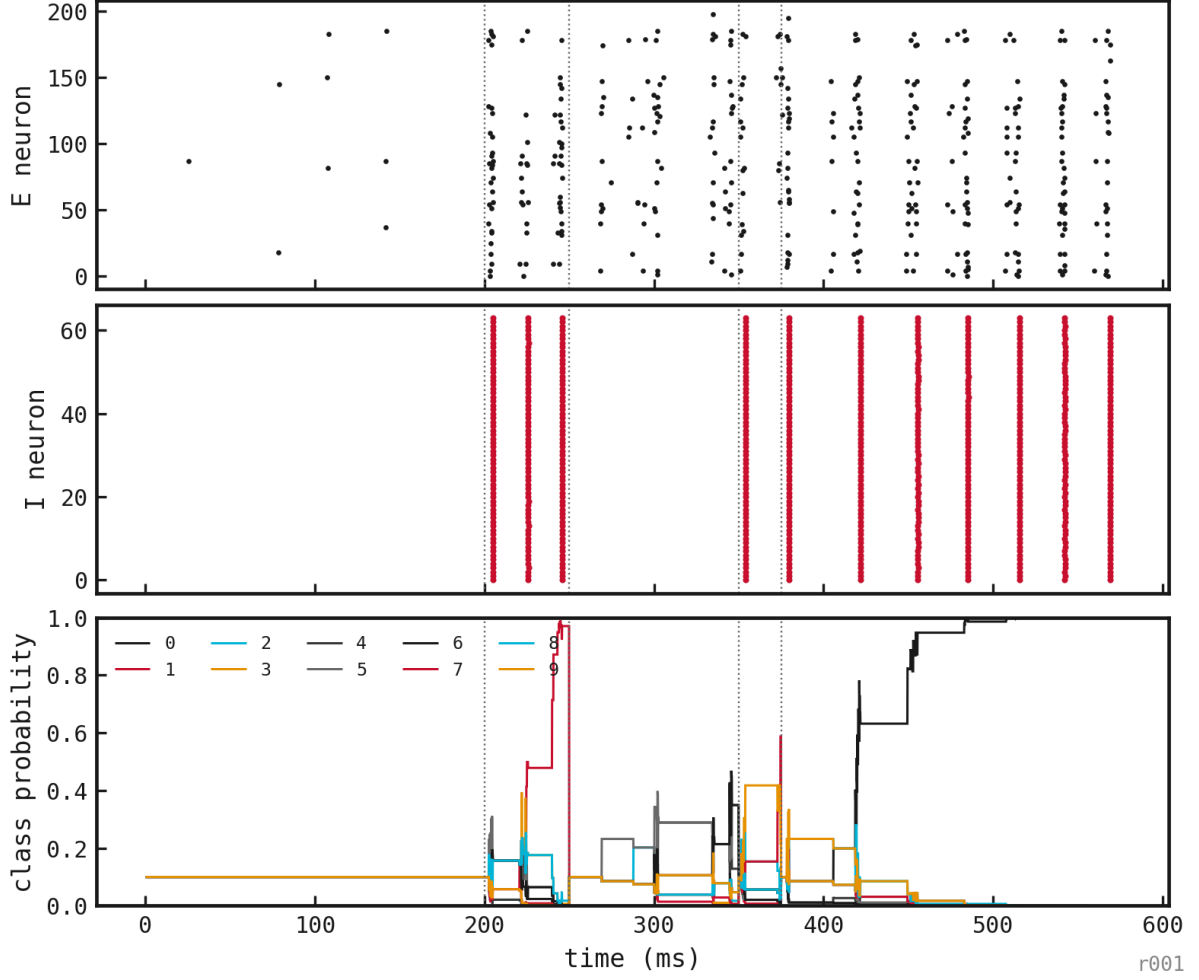


Figure 2: Streaming inference while maximum-pixel input rate and presentation duration vary between digits.

The factorial summary separates the two manipulations. The left panel crosses presentation duration with maximum-pixel input rate. The right panel holds presentation and readout at 200 ms, giving the rate psychometric without a simultaneous change in integration time.

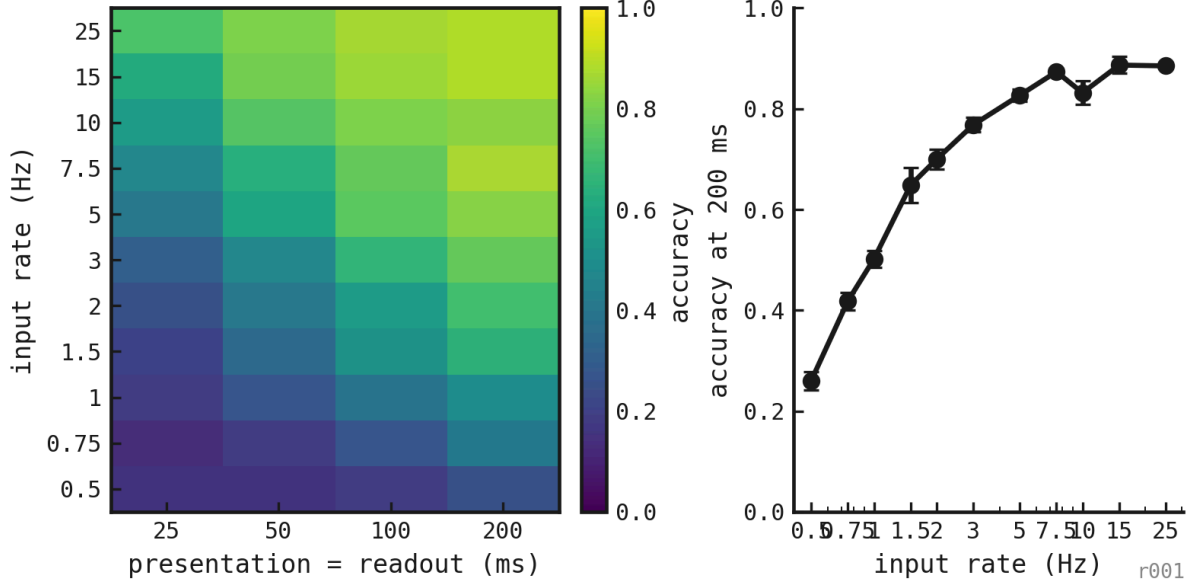


Figure 3: Accuracy across matched presentation/readout durations and input rates, with the 200 ms rate psychometric shown separately.

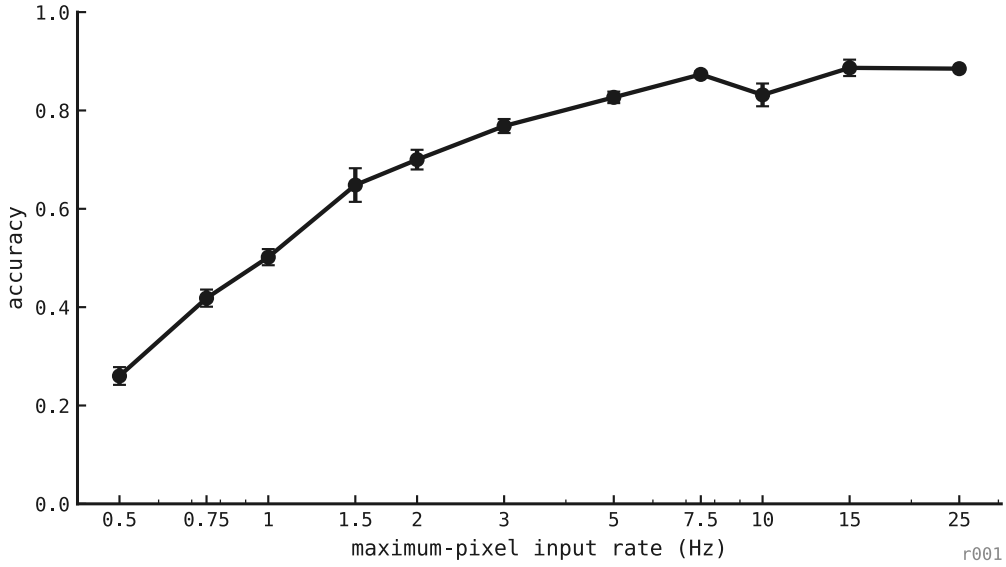


Figure 4: Input-rate psychometric at a fixed 200 ms presentation and spike-count window.

3. Comparison with exp048

Exp048 used a fixed-25-Hz training distribution and reconstructed a sliding mem-mean output. This experiment instead trains across the evaluated rates and uses the trained output-LIF spike counts directly. A numerical difference between the experiments therefore combines a change in training distribution with a change in readout; it is not an isolated comparison of either intervention.